MSG No. 191.242 P_c6/mmm [Type IV, hexagonal]

Table 1: Wyckoff site: `2a`, site symmetry: `6/mmm`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]`
2	$\left[0, 0, \frac{1}{2}\right]$	`[25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48]`

Table 2: Wyckoff site: `2b`, site symmetry: `6/m'mm`

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	`[1,2,3,4,5,6,19,20,21,22,23,24,31,32,33,34,35,36,37,38,39,40,41,42]`
2	$\left[0, 0, \frac{3}{4}\right]$	`[7,8,9,10,11,12,13,14,15,16,17,18,25,26,27,28,29,30,43,44,45,46,47,48]`

Table 3: Wyckoff site: `4c`, site symmetry: `-6m2`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	`[1,3,5,10,11,12,14,16,18,19,20,21]`
2	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[2,4,6,7,8,9,13,15,17,22,23,24]`
3	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	`[25,27,29,34,35,36,38,40,42,43,44,45]`
4	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	`[26,28,30,31,32,33,37,39,41,46,47,48]`

Table 4: Wyckoff site: `4d`, site symmetry: `-6'm2'`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{4}\right]$	`[1,3,5,19,20,21,34,35,36,38,40,42]`
2	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{4}\right]$	`[2,4,6,22,23,24,31,32,33,37,39,41]`
3	$\left[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}\right]$	`[7,8,9,13,15,17,26,28,30,46,47,48]`
4	$\left[\frac{1}{3}, \frac{2}{3}, \frac{3}{4}\right]$	`[10,11,12,14,16,18,25,27,29,43,44,45]`

Table 5: Wyckoff site: `4e`, site symmetry: `6mm`

No.	position	mapping
1	$[0, 0, z]$	`[1,2,3,4,5,6,19,20,21,22,23,24]`
2	$[0, 0, -z]$	`[7,8,9,10,11,12,13,14,15,16,17,18]`
3	$\left[0, 0, z + \frac{1}{2}\right]$	`[25,26,27,28,29,30,43,44,45,46,47,48]`
4	$\left[0, 0, \frac{1}{2} - z\right]$	`[31,32,33,34,35,36,37,38,39,40,41,42]`

Table 6: Wyckoff site: `6f`, site symmetry: `mmm`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	`[1,4,7,11,13,16,19,23]`
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[2,5,9,10,14,17,21,22]`
3	$\left[0, \frac{1}{2}, 0\right]$	`[3,6,8,12,15,18,20,24]`

continued ...

Table 6

No.	position	mapping
4	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[25,28,31,35,37,40,43,47]
5	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[26,29,33,34,38,41,45,46]
6	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[27,30,32,36,39,42,44,48]

Table 7: Wyckoff site: 6g, site symmetry: m'mm

No.	position	mapping
1	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[1,4,19,23,31,35,37,40]
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	[2,5,21,22,33,34,38,41]
3	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	[3,6,20,24,32,36,39,42]
4	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[7,11,13,16,25,28,43,47]
5	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	[8,12,15,18,27,30,44,48]
6	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	[9,10,14,17,26,29,45,46]

Table 8: Wyckoff site: 8h, site symmetry: 3m.

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	[1,3,5,19,20,21]
2	$\left[\frac{2}{3}, \frac{1}{3}, z\right]$	[2,4,6,22,23,24]
3	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	[7,8,9,13,15,17]
4	$\left[\frac{1}{3}, \frac{2}{3}, -z\right]$	[10,11,12,14,16,18]
5	$\left[\frac{1}{3}, \frac{2}{3}, z+\frac{1}{2}\right]$	[25,27,29,43,44,45]
6	$\left[\frac{2}{3}, \frac{1}{3}, z+\frac{1}{2}\right]$	[26,28,30,46,47,48]
7	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}-z\right]$	[31,32,33,37,39,41]
8	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}-z\right]$	[34,35,36,38,40,42]

Table 9: Wyckoff site: 12i, site symmetry: 2mm

No.	position	mapping
1	$\left[\frac{1}{2}, 0, z\right]$	[1,4,19,23]
2	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	[2,5,21,22]
3	$\left[0, \frac{1}{2}, z\right]$	[3,6,20,24]
4	$\left[\frac{1}{2}, 0, -z\right]$	[7,11,13,16]
5	$\left[0, \frac{1}{2}, -z\right]$	[8,12,15,18]
6	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	[9,10,14,17]
7	$\left[\frac{1}{2}, 0, z+\frac{1}{2}\right]$	[25,28,43,47]
8	$\left[\frac{1}{2}, \frac{1}{2}, z+\frac{1}{2}\right]$	[26,29,45,46]
9	$\left[0, \frac{1}{2}, z+\frac{1}{2}\right]$	[27,30,44,48]
10	$\left[\frac{1}{2}, 0, \frac{1}{2}-z\right]$	[31,35,37,40]
11	$\left[0, \frac{1}{2}, \frac{1}{2}-z\right]$	[32,36,39,42]
12	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-z\right]$	[33,34,38,41]

Table 10: Wyckoff site: `12j`, site symmetry: `m2m`

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,7,16,23]`
2	$[x,\ x,\ 0]$	`[2,10,17,21]`
3	$[0,\ x,\ 0]$	`[3,8,18,24]`
4	$[-x,\ 0,\ 0]$	`[4,11,13,19]`
5	$[-x,\ -x,\ 0]$	`[5,9,14,22]`
6	$[0,\ -x,\ 0]$	`[6,12,15,20]`
7	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[25,31,40,47]`
8	$\left[x,\ x,\ \frac{1}{2}\right]$	`[26,34,41,45]`
9	$\left[0,\ x,\ \frac{1}{2}\right]$	`[27,32,42,48]`
10	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[28,35,37,43]`
11	$\left[-x,\ -x,\ \frac{1}{2}\right]$	`[29,33,38,46]`
12	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[30,36,39,44]`

Table 11: Wyckoff site: `12k`, site symmetry: `m'2'm`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[1,23,31,40]`
2	$\left[x,\ x,\ \frac{1}{4}\right]$	`[2,21,34,41]`
3	$\left[0,\ x,\ \frac{1}{4}\right]$	`[3,24,32,42]`
4	$\left[-x,\ 0,\ \frac{1}{4}\right]$	`[4,19,35,37]`
5	$\left[-x,\ -x,\ \frac{1}{4}\right]$	`[5,22,33,38]`
6	$\left[0,\ -x,\ \frac{1}{4}\right]$	`[6,20,36,39]`
7	$\left[x,\ 0,\ \frac{3}{4}\right]$	`[7,16,25,47]`
8	$\left[0,\ x,\ \frac{3}{4}\right]$	`[8,18,27,48]`
9	$\left[-x,\ -x,\ \frac{3}{4}\right]$	`[9,14,29,46]`
10	$\left[x,\ x,\ \frac{3}{4}\right]$	`[10,17,26,45]`
11	$\left[-x,\ 0,\ \frac{3}{4}\right]$	`[11,13,28,43]`
12	$\left[0,\ -x,\ \frac{3}{4}\right]$	`[12,15,30,44]`

Table 12: Wyckoff site: `12l`, site symmetry: `mm2`

No.	position	mapping
1	$[x,\ 2x,\ 0]$	`[1,11,16,19]`
2	$[-x,\ x,\ 0]$	`[2,9,17,22]`
3	$[-2x,\ -x,\ 0]$	`[3,12,18,20]`
4	$[-x,\ -2x,\ 0]$	`[4,7,13,23]`
5	$[x,\ -x,\ 0]$	`[5,10,14,21]`
6	$[2x,\ x,\ 0]$	`[6,8,15,24]`
7	$\left[x,\ 2x,\ \frac{1}{2}\right]$	`[25,35,40,43]`
8	$\left[-x,\ x,\ \frac{1}{2}\right]$	`[26,33,41,46]`
9	$\left[-2x,\ -x,\ \frac{1}{2}\right]$	`[27,36,42,44]`
10	$\left[-x,\ -2x,\ \frac{1}{2}\right]$	`[28,31,37,47]`
11	$\left[x,\ -x,\ \frac{1}{2}\right]$	`[29,34,38,45]`
12	$\left[2x,\ x,\ \frac{1}{2}\right]$	`[30,32,39,48]`

Table 13: Wyckoff site: `12m`, site symmetry: `m'm2'`

No.	position	mapping
1	$\left[x,\ 2x,\ \frac{1}{4}\right]$	`[1,19,35,40]`
2	$\left[-x,\ x,\ \frac{1}{4}\right]$	`[2,22,33,41]`
3	$\left[-2x,\ -x,\ \frac{1}{4}\right]$	`[3,20,36,42]`
4	$\left[-x,\ -2x,\ \frac{1}{4}\right]$	`[4,23,31,37]`
5	$\left[x,\ -x,\ \frac{1}{4}\right]$	`[5,21,34,38]`
6	$\left[2x,\ x,\ \frac{1}{4}\right]$	`[6,24,32,39]`
7	$\left[-x,\ -2x,\ \frac{3}{4}\right]$	`[7,13,28,47]`
8	$\left[2x,\ x,\ \frac{3}{4}\right]$	`[8,15,30,48]`
9	$\left[-x,\ x,\ \frac{3}{4}\right]$	`[9,17,26,46]`
10	$\left[x,\ -x,\ \frac{3}{4}\right]$	`[10,14,29,45]`
11	$\left[x,\ 2x,\ \frac{3}{4}\right]$	`[11,16,25,43]`
12	$\left[-2x,\ -x,\ \frac{3}{4}\right]$	`[12,18,27,44]`

Table 14: Wyckoff site: `24n`, site symmetry: `..m`

No.	position	mapping
1	$[x,\ 0,\ z]$	`[1,23]`
2	$[x,\ x,\ z]$	`[2,21]`
3	$[0,\ x,\ z]$	`[3,24]`
4	$[-x,\ 0,\ z]$	`[4,19]`
5	$[-x,\ -x,\ z]$	`[5,22]`
6	$[0,\ -x,\ z]$	`[6,20]`
7	$[x,\ 0,\ -z]$	`[7,16]`
8	$[0,\ x,\ -z]$	`[8,18]`
9	$[-x,\ -x,\ -z]$	`[9,14]`
10	$[x,\ x,\ -z]$	`[10,17]`
11	$[-x,\ 0,\ -z]$	`[11,13]`
12	$[0,\ -x,\ -z]$	`[12,15]`
13	$\left[x,\ 0,\ z+\frac{1}{2}\right]$	`[25,47]`
14	$\left[x,\ x,\ z+\frac{1}{2}\right]$	`[26,45]`
15	$\left[0,\ x,\ z+\frac{1}{2}\right]$	`[27,48]`
16	$\left[-x,\ 0,\ z+\frac{1}{2}\right]$	`[28,43]`
17	$\left[-x,\ -x,\ z+\frac{1}{2}\right]$	`[29,46]`
18	$\left[0,\ -x,\ z+\frac{1}{2}\right]$	`[30,44]`
19	$\left[x,\ 0,\ \frac{1}{2}-z\right]$	`[31,40]`
20	$\left[0,\ x,\ \frac{1}{2}-z\right]$	`[32,42]`
21	$\left[-x,\ -x,\ \frac{1}{2}-z\right]$	`[33,38]`
22	$\left[x,\ x,\ \frac{1}{2}-z\right]$	`[34,41]`
23	$\left[-x,\ 0,\ \frac{1}{2}-z\right]$	`[35,37]`
24	$\left[0,\ -x,\ \frac{1}{2}-z\right]$	`[36,39]`

Table 15: Wyckoff site: `24o`, site symmetry: `.m.`

No.	position	mapping
1	$[x,\ 2x,\ z]$	`[1,19]`
2	$[-x,\ x,\ z]$	`[2,22]`
3	$[-2x,\ -x,\ z]$	`[3,20]`
4	$[-x,\ -2x,\ z]$	`[4,23]`
5	$[x,\ -x,\ z]$	`[5,21]`
6	$[2x,\ x,\ z]$	`[6,24]`
7	$[-x,\ -2x,\ -z]$	`[7,13]`
8	$[2x,\ x,\ -z]$	`[8,15]`
9	$[-x,\ x,\ -z]$	`[9,17]`
10	$[x,\ -x,\ -z]$	`[10,14]`
11	$[x,\ 2x,\ -z]$	`[11,16]`
12	$[-2x,\ -x,\ -z]$	`[12,18]`
13	$\left[x,\ 2x,\ z+\frac{1}{2}\right]$	`[25,43]`
14	$\left[-x,\ x,\ z+\frac{1}{2}\right]$	`[26,46]`
15	$\left[-2x,\ -x,\ z+\frac{1}{2}\right]$	`[27,44]`
16	$\left[-x,\ -2x,\ z+\frac{1}{2}\right]$	`[28,47]`
17	$\left[x,\ -x,\ z+\frac{1}{2}\right]$	`[29,45]`
18	$\left[2x,\ x,\ z+\frac{1}{2}\right]$	`[30,48]`
19	$\left[-x,\ -2x,\ \frac{1}{2}-z\right]$	`[31,37]`
20	$\left[2x,\ x,\ \frac{1}{2}-z\right]$	`[32,39]`
21	$\left[-x,\ x,\ \frac{1}{2}-z\right]$	`[33,41]`
22	$\left[x,\ -x,\ \frac{1}{2}-z\right]$	`[34,38]`
23	$\left[x,\ 2x,\ \frac{1}{2}-z\right]$	`[35,40]`
24	$\left[-2x,\ -x,\ \frac{1}{2}-z\right]$	`[36,42]`

Table 16: Wyckoff site: `24p`, site symmetry: `m..`

No.	position	mapping
1	$[x,\ y,\ 0]$	`[1,16]`
2	$[x-y,\ x,\ 0]$	`[2,17]`
3	$[-y,\ x-y,\ 0]$	`[3,18]`
4	$[-x,\ -y,\ 0]$	`[4,13]`
5	$[-x+y,\ -x,\ 0]$	`[5,14]`
6	$[y,\ -x+y,\ 0]$	`[6,15]`
7	$[x-y,\ -y,\ 0]$	`[7,23]`
8	$[y,\ x,\ 0]$	`[8,24]`
9	$[-x,\ -x+y,\ 0]$	`[9,22]`
10	$[x,\ x-y,\ 0]$	`[10,21]`
11	$[-x+y,\ y,\ 0]$	`[11,19]`
12	$[-y,\ -x,\ 0]$	`[12,20]`
13	$\left[x,\ y,\ \frac{1}{2}\right]$	`[25,40]`
14	$\left[x-y,\ x,\ \frac{1}{2}\right]$	`[26,41]`
15	$\left[-y,\ x-y,\ \frac{1}{2}\right]$	`[27,42]`
16	$\left[-x,\ -y,\ \frac{1}{2}\right]$	`[28,37]`
17	$\left[-x+y,\ -x,\ \frac{1}{2}\right]$	`[29,38]`

continued ...

Table 16

No.	position	mapping
18	$\left[y,\, -x+y,\, \frac{1}{2}\right]$	[30,39]
19	$\left[x-y,\, -y,\, \frac{1}{2}\right]$	[31,47]
20	$\left[y,\, x,\, \frac{1}{2}\right]$	[32,48]
21	$\left[-x,\, -x+y,\, \frac{1}{2}\right]$	[33,46]
22	$\left[x,\, x-y,\, \frac{1}{2}\right]$	[34,45]
23	$\left[-x+y,\, y,\, \frac{1}{2}\right]$	[35,43]
24	$\left[-y,\, -x,\, \frac{1}{2}\right]$	[36,44]

Table 17: Wyckoff site: 24q, site symmetry: m'..

No.	position	mapping
1	$\left[x,\, y,\, \frac{1}{4}\right]$	[1,40]
2	$\left[x-y,\, x,\, \frac{1}{4}\right]$	[2,41]
3	$\left[-y,\, x-y,\, \frac{1}{4}\right]$	[3,42]
4	$\left[-x,\, -y,\, \frac{1}{4}\right]$	[4,37]
5	$\left[-x+y,\, -x,\, \frac{1}{4}\right]$	[5,38]
6	$\left[y,\, -x+y,\, \frac{1}{4}\right]$	[6,39]
7	$\left[x-y,\, -y,\, \frac{3}{4}\right]$	[7,47]
8	$\left[y,\, x,\, \frac{3}{4}\right]$	[8,48]
9	$\left[-x,\, -x+y,\, \frac{3}{4}\right]$	[9,46]
10	$\left[x,\, x-y,\, \frac{3}{4}\right]$	[10,45]
11	$\left[-x+y,\, y,\, \frac{3}{4}\right]$	[11,43]
12	$\left[-y,\, -x,\, \frac{3}{4}\right]$	[12,44]
13	$\left[-x,\, -y,\, \frac{3}{4}\right]$	[13,28]
14	$\left[-x+y,\, -x,\, \frac{3}{4}\right]$	[14,29]
15	$\left[y,\, -x+y,\, \frac{3}{4}\right]$	[15,30]
16	$\left[x,\, y,\, \frac{3}{4}\right]$	[16,25]
17	$\left[x-y,\, x,\, \frac{3}{4}\right]$	[17,26]
18	$\left[-y,\, x-y,\, \frac{3}{4}\right]$	[18,27]
19	$\left[-x+y,\, y,\, \frac{1}{4}\right]$	[19,35]
20	$\left[-y,\, -x,\, \frac{1}{4}\right]$	[20,36]
21	$\left[x,\, x-y,\, \frac{1}{4}\right]$	[21,34]
22	$\left[-x,\, -x+y,\, \frac{1}{4}\right]$	[22,33]
23	$\left[x-y,\, -y,\, \frac{1}{4}\right]$	[23,31]
24	$\left[y,\, x,\, \frac{1}{4}\right]$	[24,32]

Table 18: Wyckoff site: 48r, site symmetry: 1

No.	position	mapping
1	$[x,\, y,\, z]$	[1]
2	$[x-y,\, x,\, z]$	[2]
3	$[-y,\, x-y,\, z]$	[3]
4	$[-x,\, -y,\, z]$	[4]
5	$[-x+y,\, -x,\, z]$	[5]

continued ...

Table 18

No.	position	mapping
6	$[y,\, -x+y,\, z]$	[6]
7	$[x-y,\, -y,\, -z]$	[7]
8	$[y,\, x,\, -z]$	[8]
9	$[-x,\, -x+y,\, -z]$	[9]
10	$[x,\, x-y,\, -z]$	[10]
11	$[-x+y,\, y,\, -z]$	[11]
12	$[-y,\, -x,\, -z]$	[12]
13	$[-x,\, -y,\, -z]$	[13]
14	$[-x+y,\, -x,\, -z]$	[14]
15	$[y,\, -x+y,\, -z]$	[15]
16	$[x,\, y,\, -z]$	[16]
17	$[x-y,\, x,\, -z]$	[17]
18	$[-y,\, x-y,\, -z]$	[18]
19	$[-x+y,\, y,\, z]$	[19]
20	$[-y,\, -x,\, z]$	[20]
21	$[x,\, x-y,\, z]$	[21]
22	$[-x,\, -x+y,\, z]$	[22]
23	$[x-y,\, -y,\, z]$	[23]
24	$[y,\, x,\, z]$	[24]
25	$\left[x,\, y,\, z+\frac{1}{2}\right]$	[25]
26	$\left[x-y,\, x,\, z+\frac{1}{2}\right]$	[26]
27	$\left[-y,\, x-y,\, z+\frac{1}{2}\right]$	[27]
28	$\left[-x,\, -y,\, z+\frac{1}{2}\right]$	[28]
29	$\left[-x+y,\, -x,\, z+\frac{1}{2}\right]$	[29]
30	$\left[y,\, -x+y,\, z+\frac{1}{2}\right]$	[30]
31	$\left[x-y,\, -y,\, \frac{1}{2}-z\right]$	[31]
32	$\left[y,\, x,\, \frac{1}{2}-z\right]$	[32]
33	$\left[-x,\, -x+y,\, \frac{1}{2}-z\right]$	[33]
34	$\left[x,\, x-y,\, \frac{1}{2}-z\right]$	[34]
35	$\left[-x+y,\, y,\, \frac{1}{2}-z\right]$	[35]
36	$\left[-y,\, -x,\, \frac{1}{2}-z\right]$	[36]
37	$\left[-x,\, -y,\, \frac{1}{2}-z\right]$	[37]
38	$\left[-x+y,\, -x,\, \frac{1}{2}-z\right]$	[38]
39	$\left[y,\, -x+y,\, \frac{1}{2}-z\right]$	[39]
40	$\left[x,\, y,\, \frac{1}{2}-z\right]$	[40]
41	$\left[x-y,\, x,\, \frac{1}{2}-z\right]$	[41]
42	$\left[-y,\, x-y,\, \frac{1}{2}-z\right]$	[42]
43	$\left[-x+y,\, y,\, z+\frac{1}{2}\right]$	[43]
44	$\left[-y,\, -x,\, z+\frac{1}{2}\right]$	[44]
45	$\left[x,\, x-y,\, z+\frac{1}{2}\right]$	[45]
46	$\left[-x,\, -x+y,\, z+\frac{1}{2}\right]$	[46]
47	$\left[x-y,\, -y,\, z+\frac{1}{2}\right]$	[47]
48	$\left[y,\, x,\, z+\frac{1}{2}\right]$	[48]